Deepak R

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Professional Summary

Dedicated B.E. Robotics and Automation undergraduate student (CGPA: 9.2/10.0) with practical experience across ROS 2 (Jazzy), Embodied AI (VGA Policy / LeRobot), Autonomous Navigation (Nav2, SLAM), Computer Vision, and Embedded RTOS Systems. Demonstrated capabilities across multi-sensor fusion, edge AI deployment, and industrial robotics through internships, hackathons, open-source contributions, and peer-reviewed preprint research.

Education

Dhaanish Ahmed College of Engineering (**Anna University**) **Chennai, India**
Bachelor of Engineering (B.E.) in Robotics and Automation – CGPA: 9.2/10.0 *May 2024 – May 2028*

- **Relevant Coursework:** Industrial Robotics, Autonomous Mobile Robots, Control Systems, Digital Circuits, Machine Learning, Embedded Systems, ROS 2 Navigation, Kinematics & Dynamics, Computer Integrated Manufacturing.

Work Experience

Wildplant Terrestrial Solutions Pvt Ltd – *Web Development Intern (Under NDA)* **Remote | Jul 2026 – Aug 2026**

- Architected full-stack platforms and secure RESTful backend APIs in Python/Node.js for proprietary internal systems under NDA.
- Optimized client-side data flows and integrated real-time telemetry streaming for internal developer dashboards.

Precise3DM – *Automation & Data Engineering Intern* **Chennai, India (Hybrid) | Jun 2026 – Jul 2026**

- Developed automated ETL data processing pipelines and web scrapers integrating REST API services, eliminating manual ingestion.
- Built structured validation logic and automated logging routines to ensure high pipeline reliability and data integrity.

Tamizhan Skills (RISE Programs) – *AI & Autonomous Systems Intern* **Remote | Dec 2025 – Mar 2026**

- Configured ROS 2 Nav2 navigation stacks, costmaps, and multi-sensor fusion (LiDAR, IMU, Odometry) in Gazebo simulations.
- Trained CNN models in TensorFlow/Keras for image classification and engineered NLP text anomaly detection classifiers.

Chennai Port Authority – *Industrial Engineering Intern* **Chennai, India | Dec 2025**

- Diagnosed and serviced diesel locomotive powertrains; analyzed PLC-driven port logistics and automation systems.

Publications

"AURA: Resilient Urban Traffic Signal Control via Semi-Markov RL" **IEEE TITS Under Review**
Zenodo: 10.5281/zenodo.20265628 | IEEE TITS (Submission ID dbb571bd) *Nov 2026*

- Formulated SMDP-RL with action-duration bounds and shockwave-mitigated emergency preemption (42% faster dissipation).
- Authored AutoTwin-AI chassis digital twin; submitted NeuroCode to Journal of Parallel and Distributed Computing.

Projects

VGA Policy: Continuous 3D Ray-RoPE Robotic Manipulation – *PyTorch, LeRobot, MuJoCo* **RSS 2027 Submission**

- Developed 375M VLA policy with continuous 3D Ray-RoPE and velocity-matching loss (L_{smooth}); achieved **85.0% success** on LIBERO-Spatial, **35.4% jerk reduction**, and 74.5% success under camera extrinsics shift in LIBERO-MAX.

NeuroCode: Heterogeneous Code-Graph Slicing Framework – *Python, MCP, Tree-sitter* **JPDC Under Review**

- Published **neurocode-ai v1.4.0** on PyPI with MCP server for local LLMs; resolved SWE-bench Lite on 4GB VRAM laptop GPUs (0 OOMs), 100% recall on RepoBench-R; submitted to Journal of Parallel and Distributed Computing (JPDC-S-26-02633).

AutoTwin-AI: Automotive Chassis Digital Twin & Anomaly Detection – *PyTorch, Blender Cycles OptiX, Three.js* **Sep 2026**

- Engineered an AI digital twin for vehicle ladder-frame chassis assemblies; trained PyTorch convolutional autoencoders on 5,000+ Blender OptiX synthetic CAD images to detect weld defects via $|I - \hat{I}|$ residual heatmaps.

WeldGuard-AI: In-Situ Universal Robotic Welding Surveillance – *PyTorch, ConvLSTM, OpenCV* **Oct 2026**

- Built edge welding surveillance with ambient glare exclusion (T_0 calibration), adaptive torch-tip tracking (400×400 ROI yielding 13.6x speedup), Weld Pool Geometry Monitoring (WPGM), and zero-flapping safety E-STOP.

Technical Skills

- **Languages:** Python, C++, C, SQL, Rust, Bash, JavaScript, HTML/CSS.
- **Robotics & Simulation:** ROS 2 (Jazzy), Gazebo Harmonic, Nav2 Stack, SLAM Toolbox, MoveIt 2, MuJoCo Physics, LeRobot, ABB RobotStudio & RAPID, URDF/Xacro, Path Planning (RRT/PRM/A*), Coverage Planning (BCD).
- **AI, CV & Web:** PyTorch, Diffusion Transformers (DiT), Flow Matching, Model Context Protocol (MCP), OpenCV, YOLO (v8/v11), MediaPipe, TensorFlow, Scikit-learn, FastAPI, Node.js, Next.js.
- **Embedded Systems & Tools:** STM32, Raspberry Pi 4, ESP32, ESP8266, Arduino, Linux (Ubuntu 24.04/WSL2), Docker, Git, Fusion 360, LaTeX, Hugging Face (@Arkz121), OpenReview (~Deepak_R2).